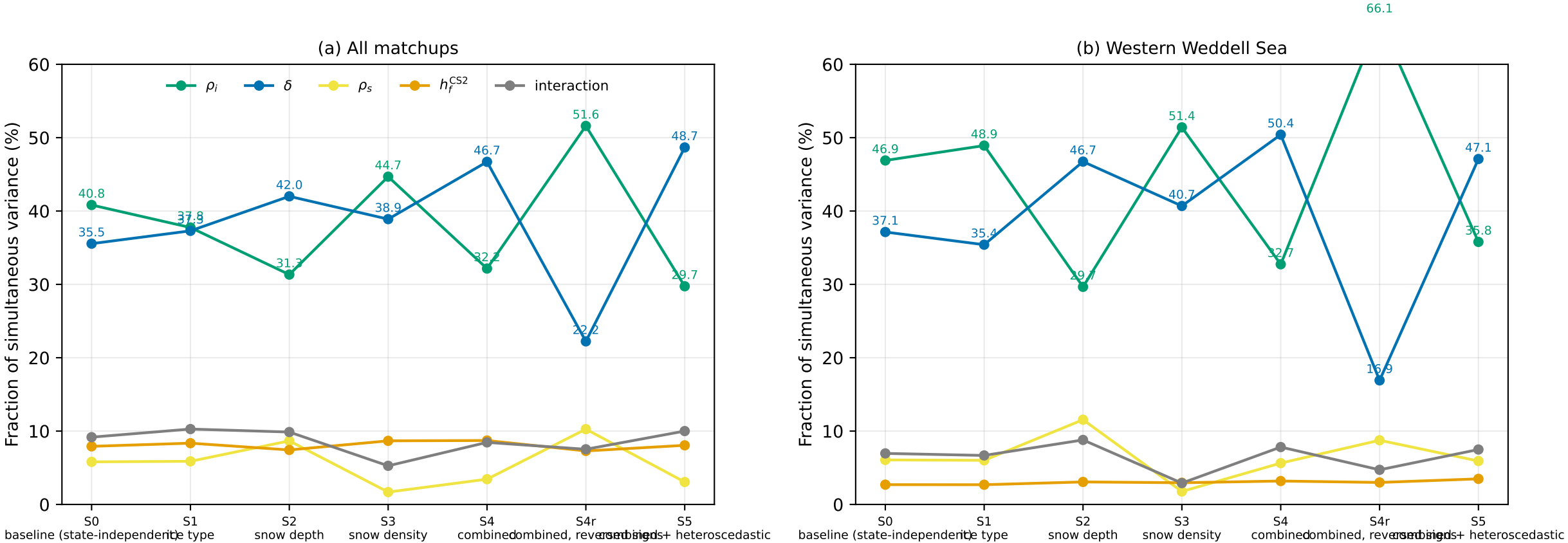Variance budget with the penetration-fraction prior conditioned on ice type, snow depth and snow density